Confusion Matrix

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

AI-2 Week 13: define positive before naming errors.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Build a labelled confusion matrix from matched reference and prediction pairs. Try the recall before opening notes.

Your learning target

I can build a labelled **confusion matrix** from matched reference and **prediction** pairs.

I can distinguish **false positive**s from **false negative**s using a declared **positive class**.

I can check matrix totals and explain why equal accuracy can hide different errors.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

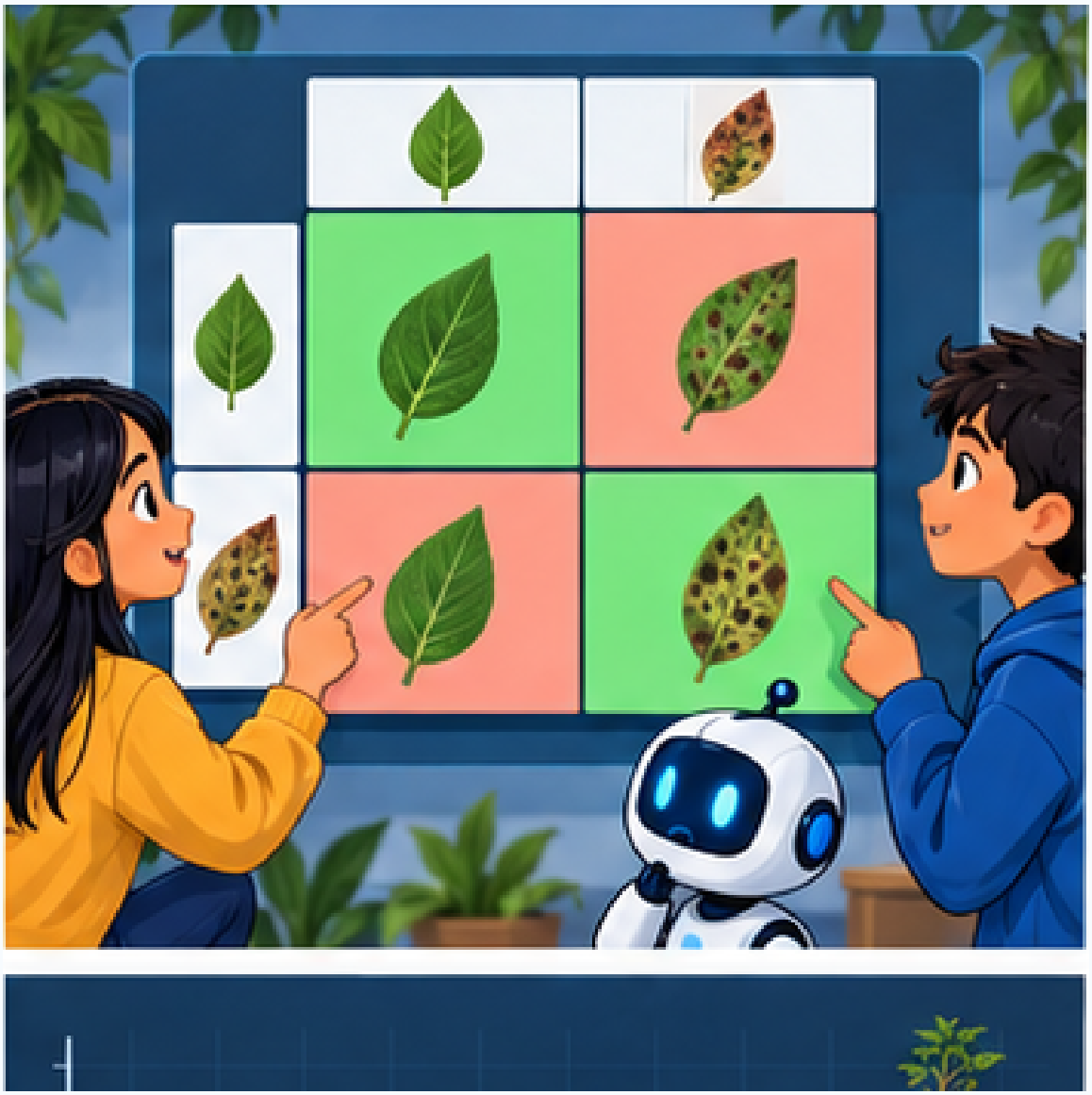
Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A **confusion matrix** separates **true positive**s, **false positive**s, **false negative**s, and **true negative**s so error types remain visible.



Your target

LEARNING TARGET Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

Builder name and experiment date:

Readiness for the connected explanation

Why should a reproducible run save **prediction**s linked to example IDs rather than only an accuracy score?

This week's end goal

By the end, I can explain **confusion matrix** using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

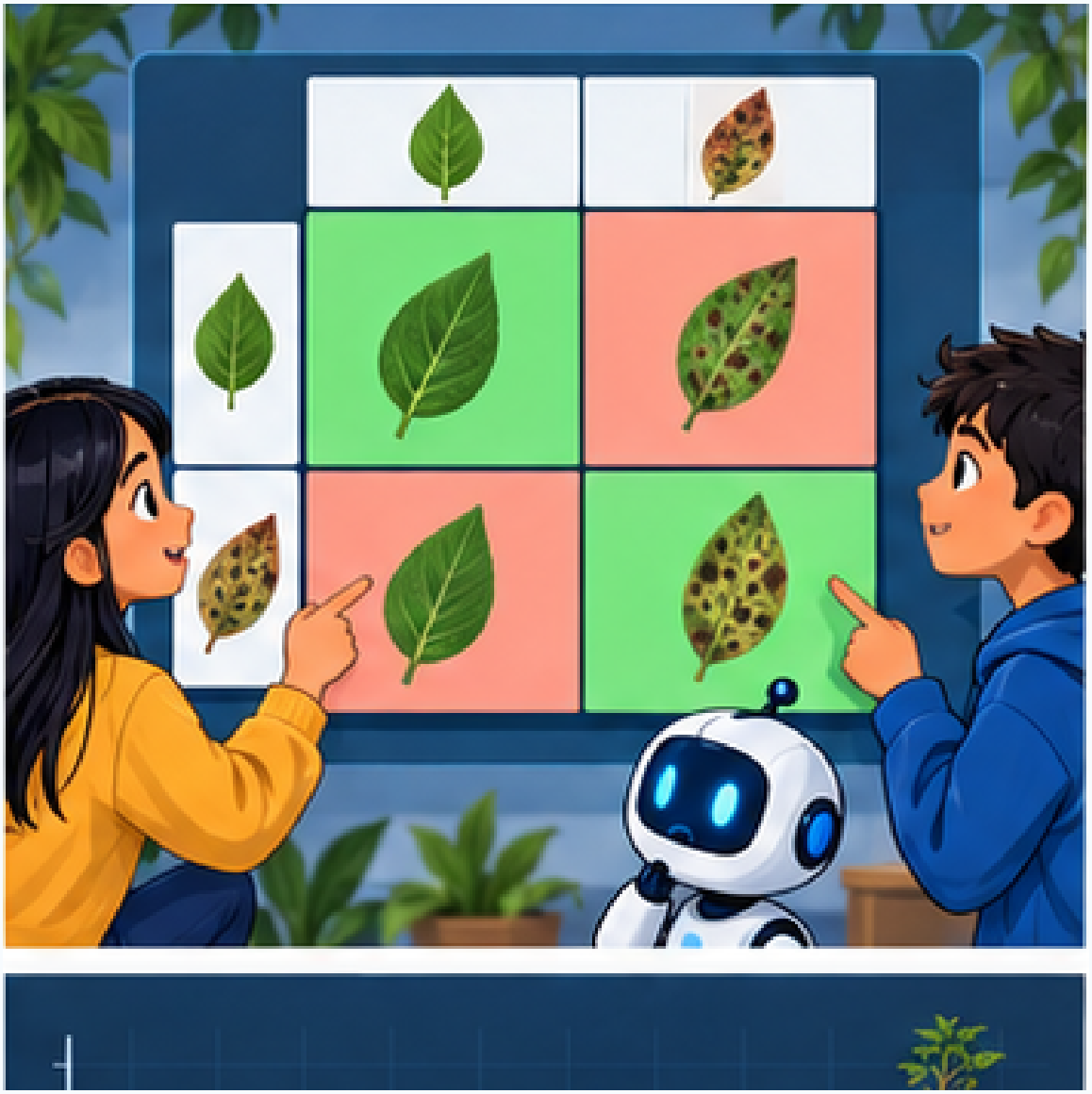
Notice the evidence

> Investigate the supplied Confusion Matrix case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **confusion matrix** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A **confusion matrix** shows which labels a classifier mixes up. It counts examples after **prediction**s are made; it does not fit or repair the classifier.

Week 12 had reference/**prediction** pairs clear/clear, blocked/blocked, clear/blocked. A labelled table can preserve the type of that error as well as the total score.

This week uses eight new fictional evaluation records from a frozen classifier. They are not extra **prediction**s from the Week 12 code. Each ID appears once with a reviewed reference and one **prediction**.

The grouped table below lists every ID. Keep reference and **prediction** paired by ID before counting.

Evaluation IDs	Reference	Prediction
V1 V2	clear	clear
V3	clear	blocked
V4 V5	blocked	clear
V6 V7 V8	blocked	blocked

Use reference labels for rows and predicted labels for columns. On both axes the order is clear, then blocked. Declare blocked as the **positive class** and clear as negative.

Positive means the condition of interest, not good or safe. A missed blocked case and a false alarm are different errors.

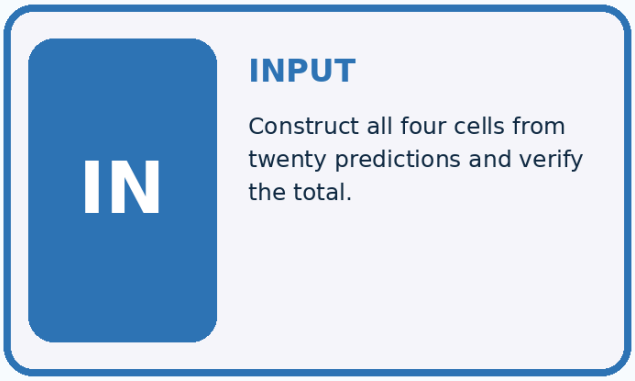
See how it works

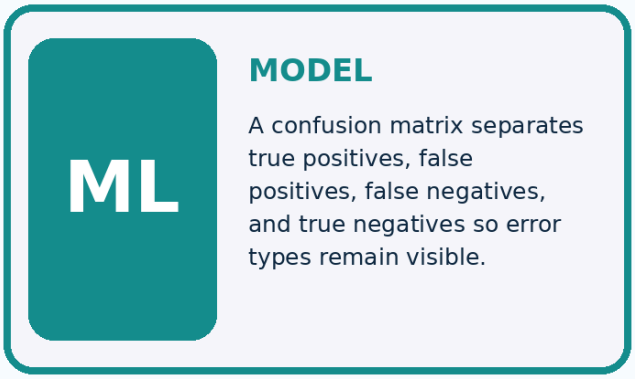
> Follow the mechanism for Confusion Matrix. Connect each labelled input, deciding step and result.

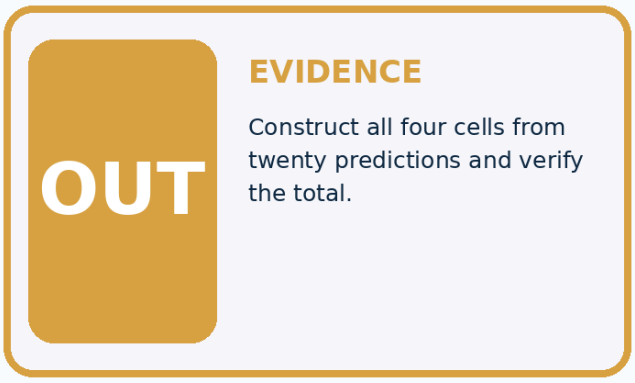
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Construct all four cells from twenty **prediction**s and verify the total.

MODEL IDEA A **confusion matrix** separates **true positive**s, **false positive**s, **false negative**s, and **true negative**s so error types remain visible.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD For spotted=positive: TP=6, FP=3, FN=2, TN=9. Check total: 6 + 3 + 2 + 9 = 20 cases.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

```
pairs = [("V1", "clear", "clear"),
         ("V2", "clear", "clear"),
         ("V3", "clear", "blocked"),
         ("V4", "blocked", "clear"),
         ("V5", "blocked", "clear"),
         ("V6", "blocked", "blocked"),
         ("V7", "blocked", "blocked"),
         ("V8", "blocked", "blocked")]
```

```
labels = ["clear", "blocked"]
index = {label: i for i, label in enumerate(labels)}
counts = [[0, 0], [0, 0]]
for item_id, reference, prediction in pairs:
    counts[index[reference]][index[prediction]] += 1
print(counts)
correct = counts[0][0] + counts[1][1]
print(correct, len(pairs))
```

index maps clear to 0 and blocked to 1. Each paired record adds one to its reference row and **prediction** column. The script prints [[2, 1], [2, 3]] then 5 8. It assumes unique IDs and only these two labels.

Check IDs and missing or unresolved references before counting; the code does not validate all those conditions.

Reference rows	Predicted clear	Predicted blocked
Reference clear	2	1
Reference blocked	2	3

The diagonal cells count correct **prediction**s: 2+3=5. Off-diagonal cells count mistakes: 1+2=3. Every record must contribute exactly once. This is a count matrix, not percentages or probabilities.

Keep the axis labels even when copying the numbers.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

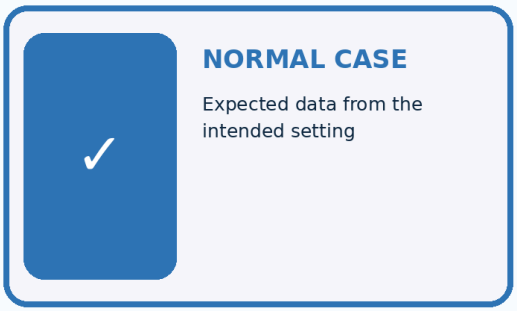
Compare the cases

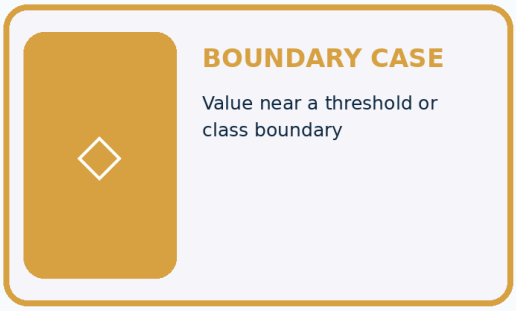
> Compare the supplied cases for Confusion Matrix. Identify the detail that changes the conclusion.

EVALUATION DESIGN

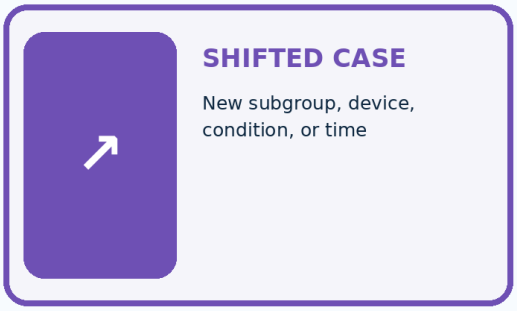
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: A review corrects one missed spotted leaf, moving it from FN to TP.

Predicted result

Prediction: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

Why the result follows

Reason: One decision becomes correct while the total number of evaluated leaves remains20.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Confusion Matrix to explain a concrete decision, rather than only memorising definitions.

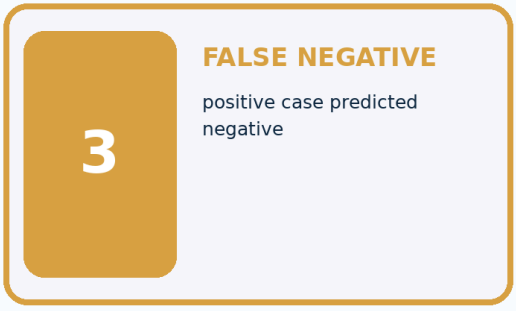
VOCABULARY

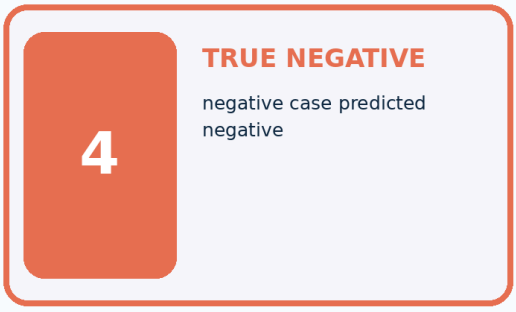
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
true positive	positive case predicted positive	
false positive	negative case predicted positive	
false negative	positive case predicted negative	
true negative	negative case predicted negative	

MISCONCEPTION TO REPAIR The diagonal alone tells everything important about a classifier.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Confusion matrix	A table counting reference and prediction combinations
Positive class	The class chosen as the condition of interest
False positive	A positive prediction when the reference is negative
False negative	A negative prediction when the reference is positive

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Confusion Matrix, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 15 • WORKED EVIDENCE

1. For spotted=positive: TP=6, FP=3, FN=2, TN=9.
2. Check total: 6 + 3 + 2 + 9 = 20 cases.
3. Correct predictions = TP + TN = 15; accuracy = 15 ÷ 20 = 75%.
4. Inspect false alarms and misses separately before choosing an improvement.

Step 1

STEP 1 For spotted=positive: TP=6, FP=3, FN=2, TN=9.

Step 2

STEP 2 Check total: 6 + 3 + 2 + 9 = 20 cases.

Step 3

STEP 3 Correct **prediction**s = TP + TN = 15; accuracy = 15 ÷ 20 = 75%.

Step 4

STEP 4 Inspect false alarms and misses separately before choosing an improvement.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Blocked is the positive class

REFERENCE AND PREDICTION	CELL NAME	IDS AND COUNT
clear then clear	True negative TN	V1 V2 count 2
clear then blocked	False positive FP	V3 count 1
blocked then clear	False negative FN	V4 V5 count 2
blocked then blocked	True positive TP	V6 V7 V8 count 3

Read reference first and prediction second. Here a false positive is a false blocked alarm; a false negative is a missed blocked case.

Place V3 and V4 in the matrix. Explain which is a false positive and which is a false negative when blocked is positive.

True means **prediction** agrees with the reference. It does not prove that the reference annotation is infallible. Resolve disputed labels with the documented review policy rather than silently changing them to match **prediction**s.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Confusion Matrix

Read the **confusion matrix** as four distinct stories about a plant decision.

The facts we will use

Supplied facts

Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/15

Positive means spotted

	predicted spotted	predicted healthy
actual spotted	4	1
actual healthy	3	12

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/15

Confusion Matrix

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fill actual spotted row

Actual spotted: [4, 1]. Four caught; one missed.

2. Fill actual healthy row

Actual healthy: [3, 12]. Three false alarms; twelve correctly clear.

3. Compute accuracy

Correct=4+12=16 of total 20, giving80%.

CHANGE ONE THING

A review corrects one missed spotted leaf, moving it from FN to TP.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fill actual spotted row	Actual spotted: [4, 1]. Four caught; one missed.
2. Fill actual healthy row	Actual healthy: [3, 12]. Three false alarms; twelve correctly clear.
3. Compute accuracy	Correct=4+12=16 of total 20, giving80%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Correct=4+12=16 of total 20, giving**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

The matrix depends on positive-class definition and reference labels. Accuracy alone hides the difference between misses and false alarms.

Keep the stated limit

Read the labelled picture: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: A review corrects one missed spotted leaf, moving it from FN to TP. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Confusion Matrix claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The diagonal alone tells everything important about a classifier. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE The diagonal alone tells everything important about a classifier.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Check the claim against the facts

Always report the evaluated split, number of records, reference and **prediction** axes, label order and **positive class**. Reversing the axis order changes positions, not the underlying outcomes.

Transposing rows and columns swaps the two kinds of error unless the labels are corrected. Read the headers before interpreting a diagram from another source.

Check the claim against the facts

A zero cell means no such event in this evaluated set, not that it is impossible. A **confusion matrix** does not show the cause of errors, label quality or performance on future data. Keep the ID-level records for investigation.

Next week examines accuracy when class counts are unequal.

Two classifiers have accuracy 5/8. One misses two blocked records and the other misses three. Explain how both statements can be true and what must accompany their matrices for a fair comparison.

STUDENT LESSON · 8 OF 18

Practise together

Practise Confusion Matrix with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + **positive class** + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Row totals are 3 reference-clear and 5 reference-blocked. Column totals are 4 predicted-clear and 4 predicted-blocked. All four cells sum to 8. Accuracy is the diagonal total divided by the total: 5/8, or 62.5 percent.

The Week 12 three-record matrix was [[1,1],[0,1]] in this same axis order, giving 2/3.

Another classifier on these same eight reference-labelled records could have matrix [[3,0],[3,2]]. It also gets 5/8 correct, but has zero **false positive**s and three **false negative**s.

Our worked classifier has one **false positive** and two **false negative**s. Equal accuracy does not settle which error pattern is preferable; consider the task and **evidence** before choosing.

Practise with less help: Explain the two error types

Use workbook W2 for the connected example “Confusion matrices”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

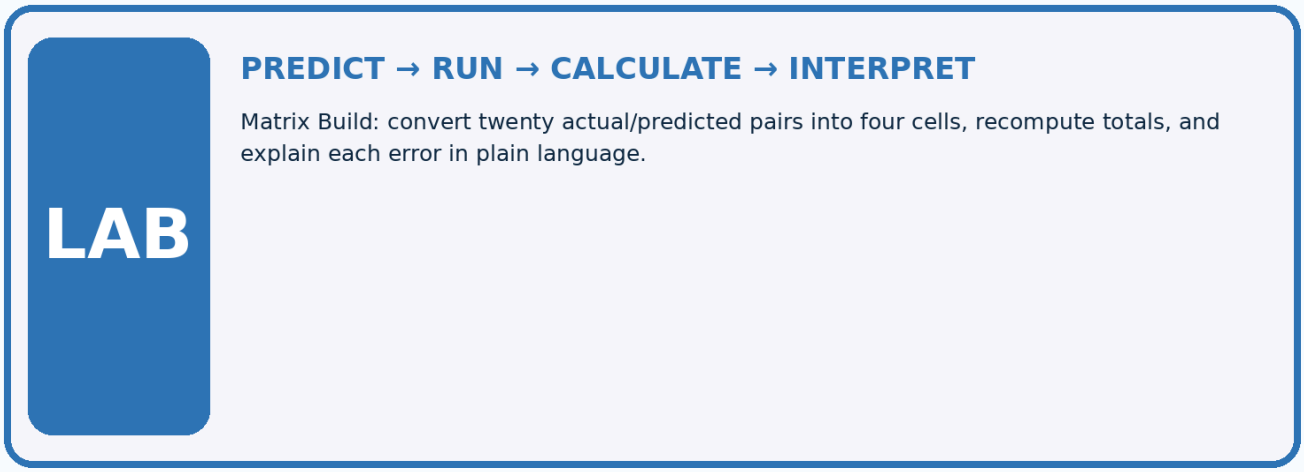
Run the investigation

> Investigate Confusion Matrix. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Matrix Build: convert twenty actual/predicted pairs into four cells, recompute totals, and

explain each error in plain language.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Boundary
Prediction: _____
Actual: _____
Calculation / evidence: _____
Meaning / next test: _____

Case: Missing
Prediction: _____
Actual: _____
Calculation / evidence: _____

> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

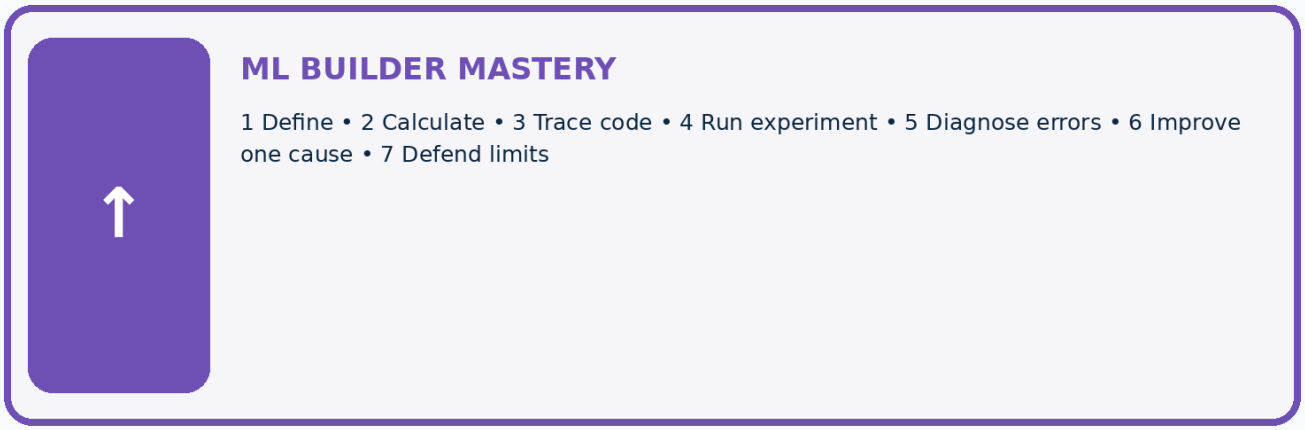
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Confusion Matrix. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **true positive**, **false positive**, **false negative**, **true negative**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The diagonal alone tells everything important about a classifier.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

STUDENT LESSON · 11 OF 18

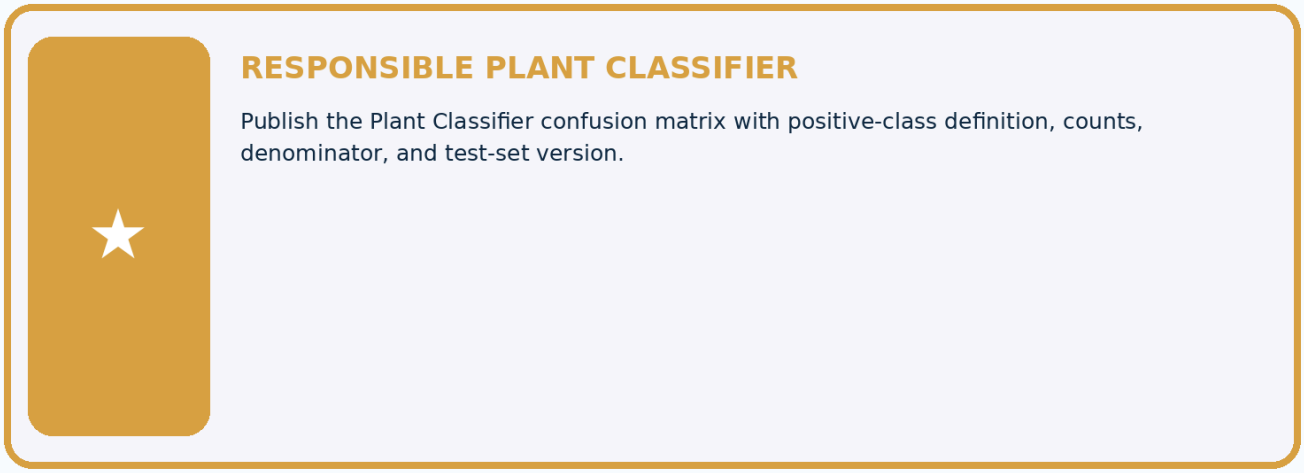
Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Publish the Plant Classifier **confusion matrix** with positive-class definition, counts, denominator, and test-set version.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

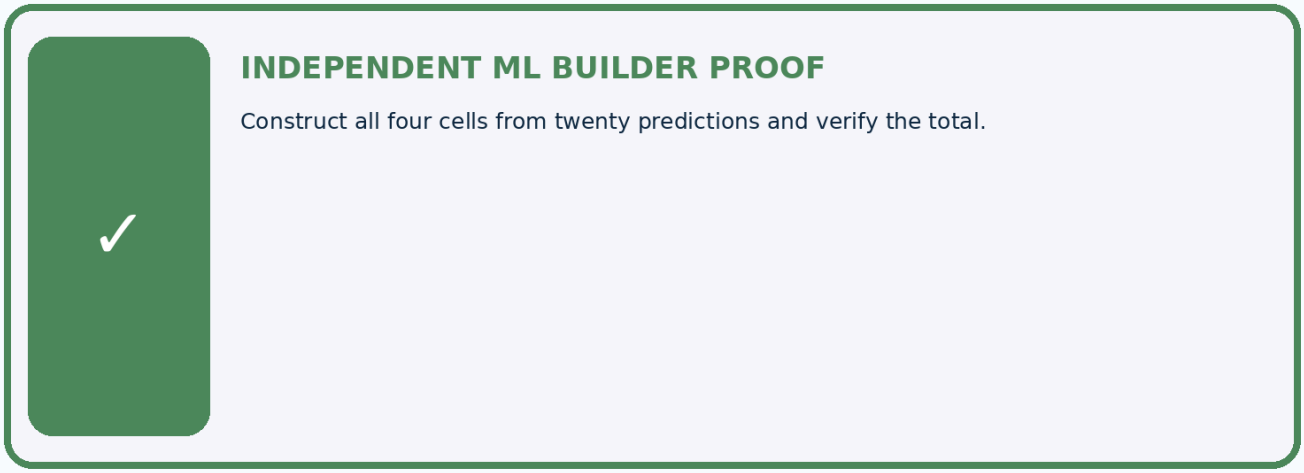
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Construct all four cells from twenty **prediction**s and verify the total.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

> **Connect to the next lesson**
>
> NEXT BRIDGE Carry the experiment **evidence** into: Accuracy and Imbalance.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Confusion Matrix case for supported practice. Repeated examples do not become fresh assessments.

Week 15 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Positive is spotted. TP=8, FP=2, FN=4, TN=6.

The question

Build actual-row/predicted-column matrix and total.

Worked explanation

Actual spotted row is 8 spotted,4 healthy; actual healthy row is 2 spotted,6 healthy. Total=20; correct=14. State axes and **positive class**.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Actual rows, predicted columns

Evidence or stage	Value or result
True positive	8
False negative	4
False positive	2
True negative	6

Positive is spotted. TP=8, FP=2, FN=4, TN=6.

Build actual-row/predicted-column matrix and total.

Actual spotted row is 8 spotted,4 healthy; actual healthy row is 2 spotted,6 healthy. Total=20; correct=14. State axes and **positive class**.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

On20 cases: **true positive**s5, **false negative**s4, **false positive**s2, **true negative**s9. Build the **confusion matrix** with actual class as rows and predicted class as columns. Compute accuracy.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 15 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> Treat fern as positive. Counts are TP=6, FP=2, FN=3 and TN=9. How many examples and how many correct **prediction**s are there?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Publish the Plant Classifier **confusion matrix** with positive-class definition, counts, denominator, and test-set version.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Positive=spotted. TP=6,FN=2,FP=3,TN=9. Draw a matrix with actual labels in rows and **prediction**s in columns. Compute total and accuracy.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Confusion Matrix, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Read the **confusion matrix** as four distinct stories about a plant decision.

By the end, I can	How I will show it
Build a labelled confusion matrix from matched reference and prediction pairs	Explain the deciding step, show a trace or test, and state one limit.
Distinguish false positives from false negatives using a declared positive class	Explain the deciding step, show a trace or test, and state one limit.
Check matrix totals and explain why equal accuracy can hide different errors	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Confusion Matrix without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Build a labelled **confusion matrix** from matched reference and **prediction** pairs

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read the **confusion matrix** as four distinct stories about a plant decision.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Correct=4+12=16 of total 20, giving**80%**.

1. Compute accuracy

2. Fill actual spotted row

3. Fill actual healthy row

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compute accuracy Correct=4+12=16 of total 20, giving**80%**. Read the picture in order. Correct=4+12=16 of total 20, giving**80%**.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: A review corrects one missed spotted leaf, moving it from FN to TP. Which conclusion is supported by the supplied facts?

1. Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%. One decision becomes correct while the total number of evaluated leaves remains20.

Repair a missing building block

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score mean every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset Cards (ai-4/9). Goal: Inspect row meaning missing values and repeated records. Short practice: What two counts should the revised card show?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

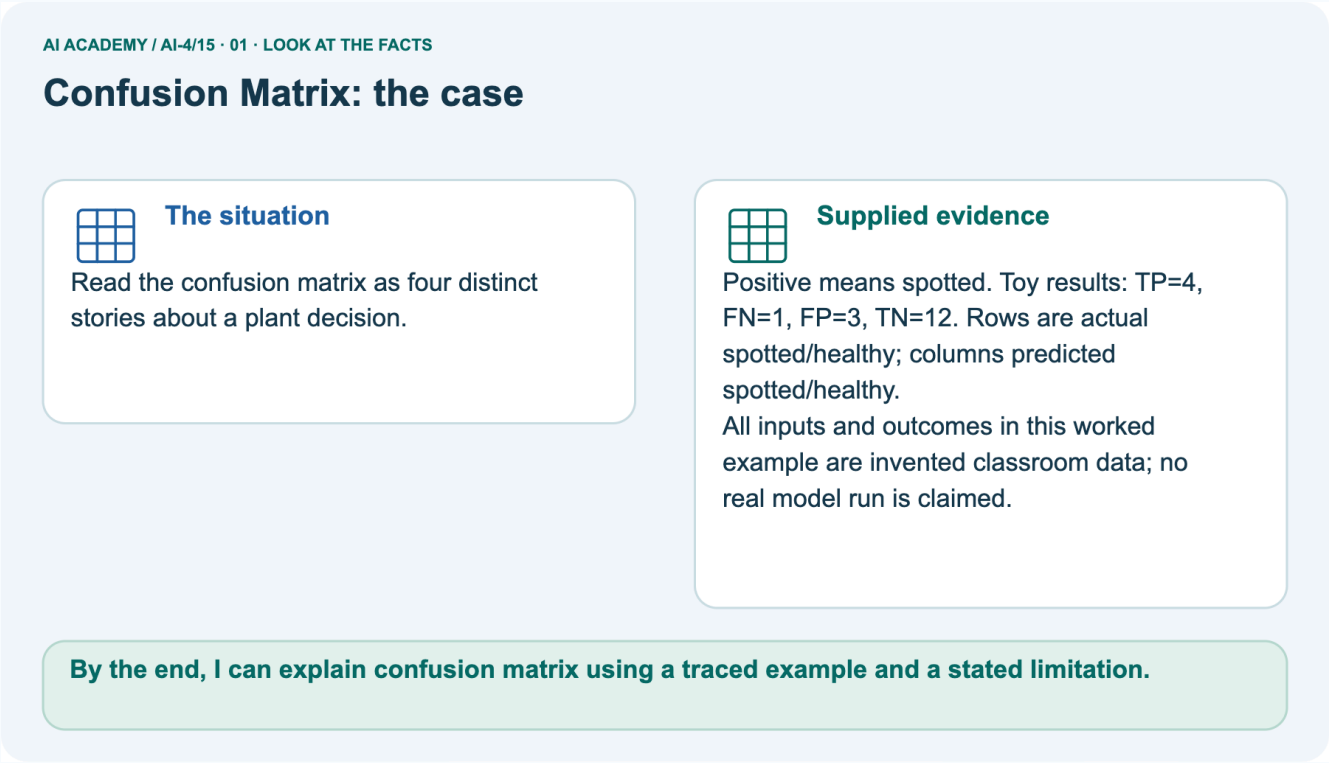
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Confusion Matrix graphic using its labels, deciding evidence and relevant limitation.

01. Confusion Matrix: the case



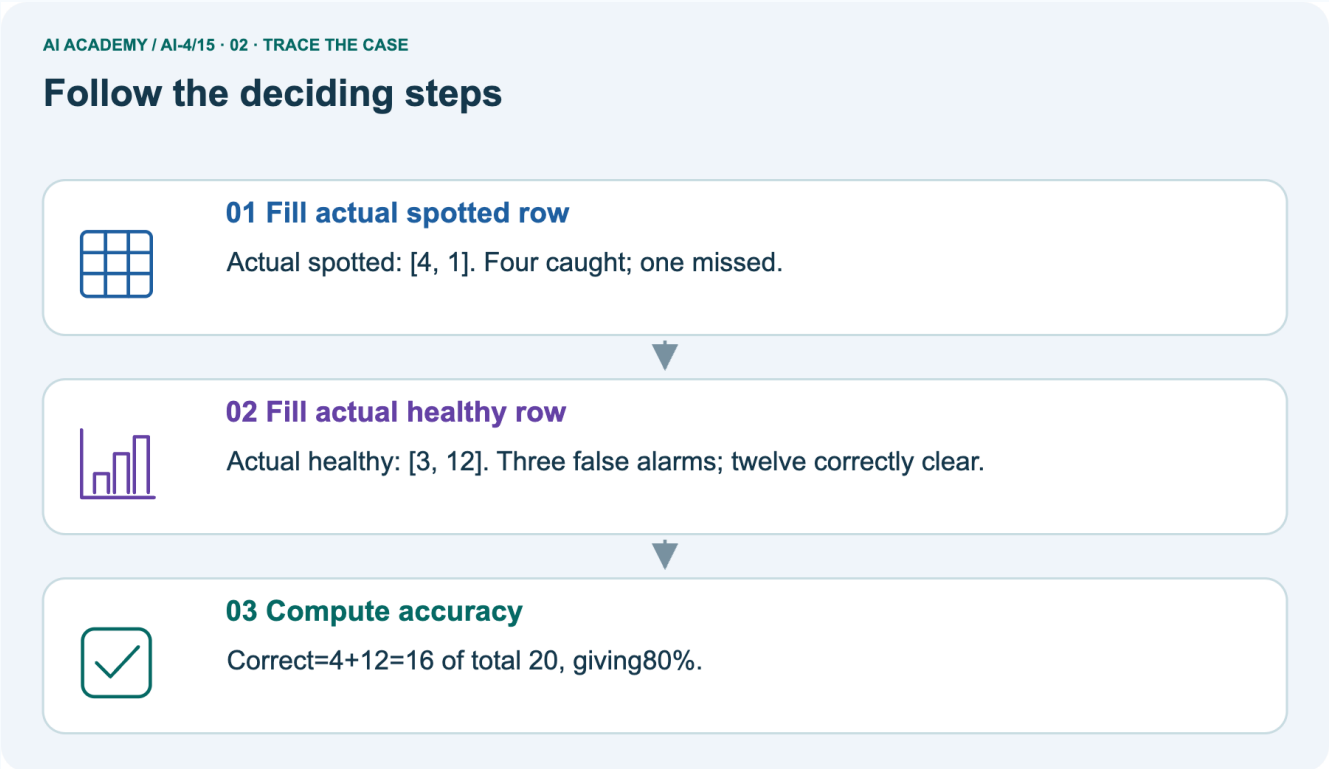
Follow the labelled picture with the matching explanation. Open the vector diagram

Read the **confusion matrix** as four distinct stories about a plant decision. Supplied facts: Positive means spotted. Toy results: TP=4, FN=1, FP=3, TN=12. Rows are actual spotted/healthy; columns predicted spotted/healthy.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



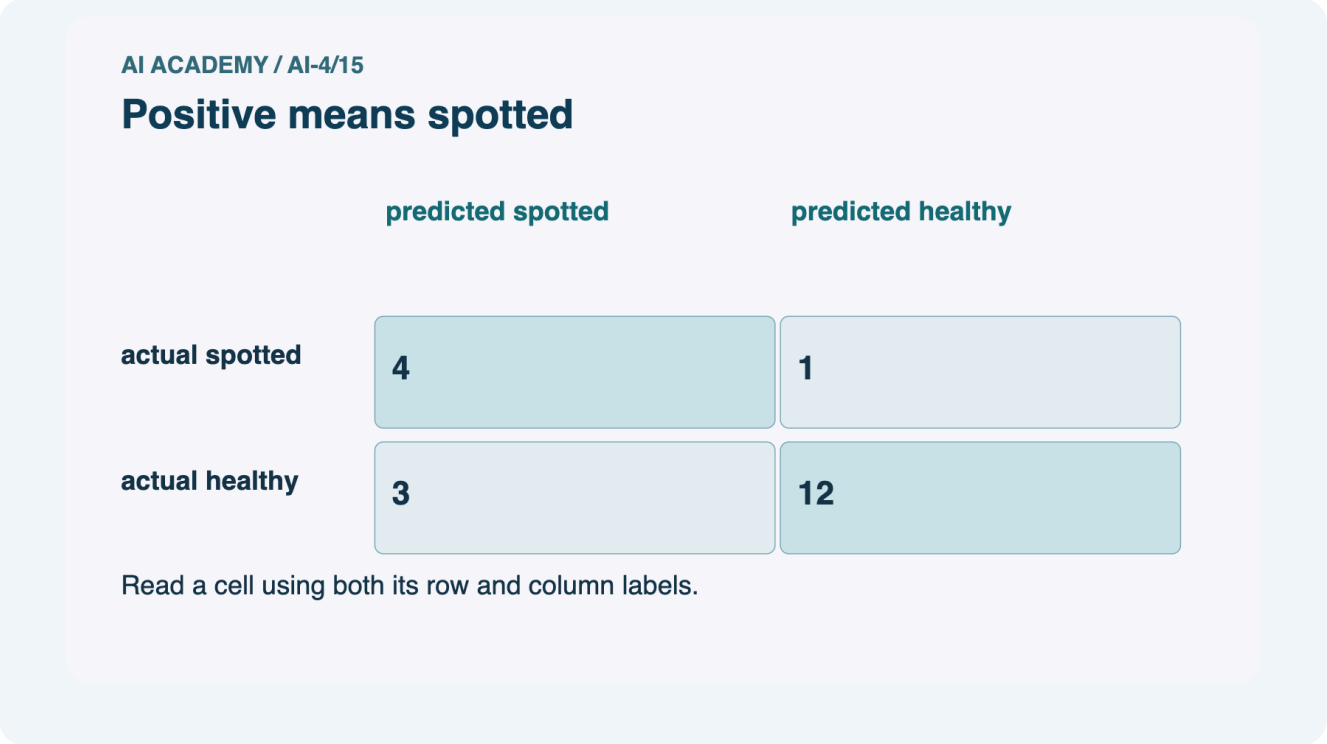
Follow the labelled picture with the matching explanation. Open the vector diagram

Fill actual spotted row: Actual spotted: [4, 1]. Four caught; one missed. → Fill actual healthy row: Actual healthy: [3, 12].

Three false alarms; twelve correctly clear. → Compute accuracy: Correct=4+12=16 of total 20, giving**80%**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Positive means spotted

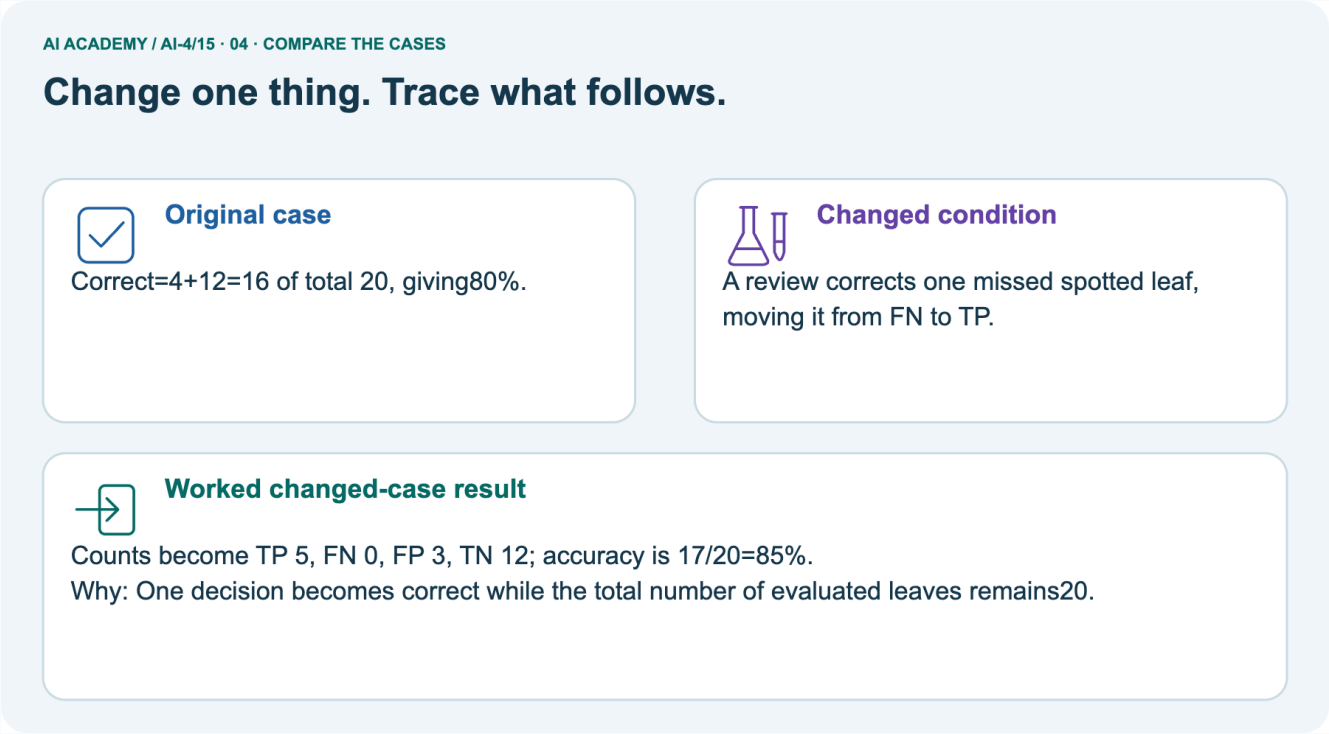


Follow the labelled picture with the matching explanation. Open the vector diagram

Positive means spotted. actual spotted: predicted spotted = 4, predicted healthy = 1. actual healthy: predicted spotted = 3, predicted healthy = 12.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



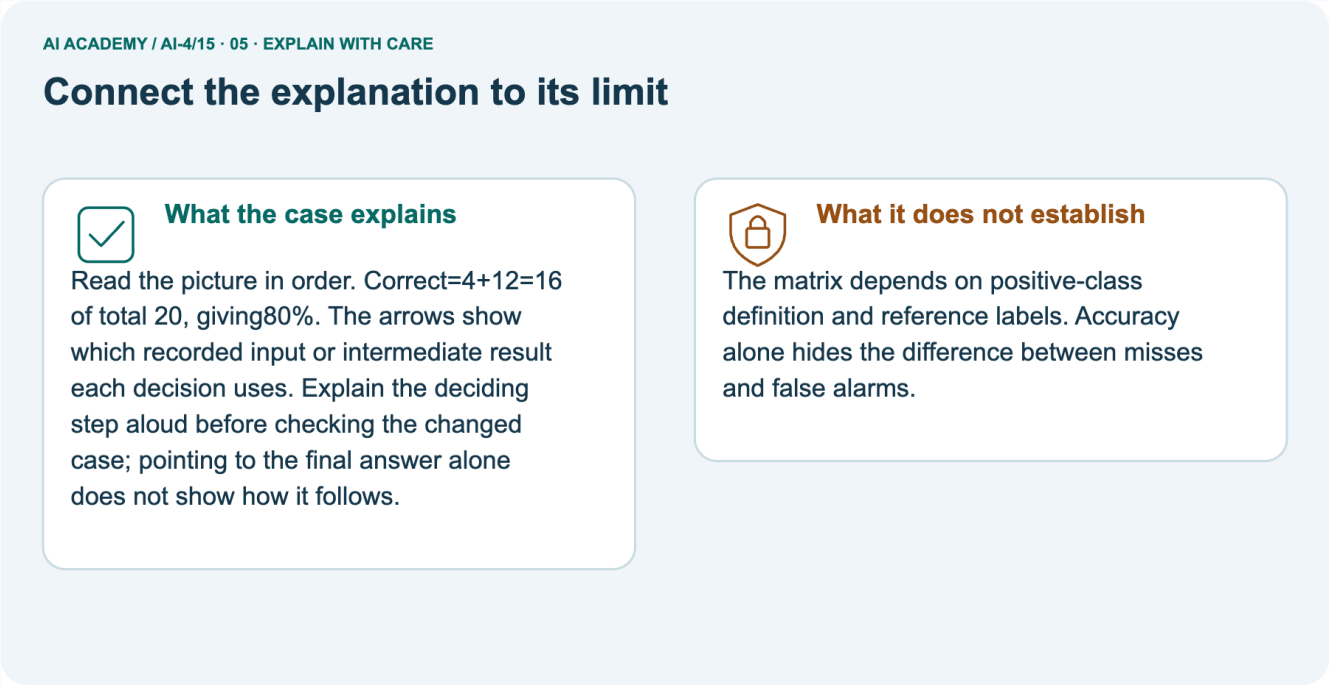
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Correct=4+12=16 of total 20, giving**80%**. Changed condition: A review corrects one missed spotted leaf, moving it from FN to TP. Expected result: Counts become TP 5, FN 0, FP 3, TN 12; accuracy is 17/20=85%.

Reason: One decision becomes correct while the total number of evaluated leaves remains20.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

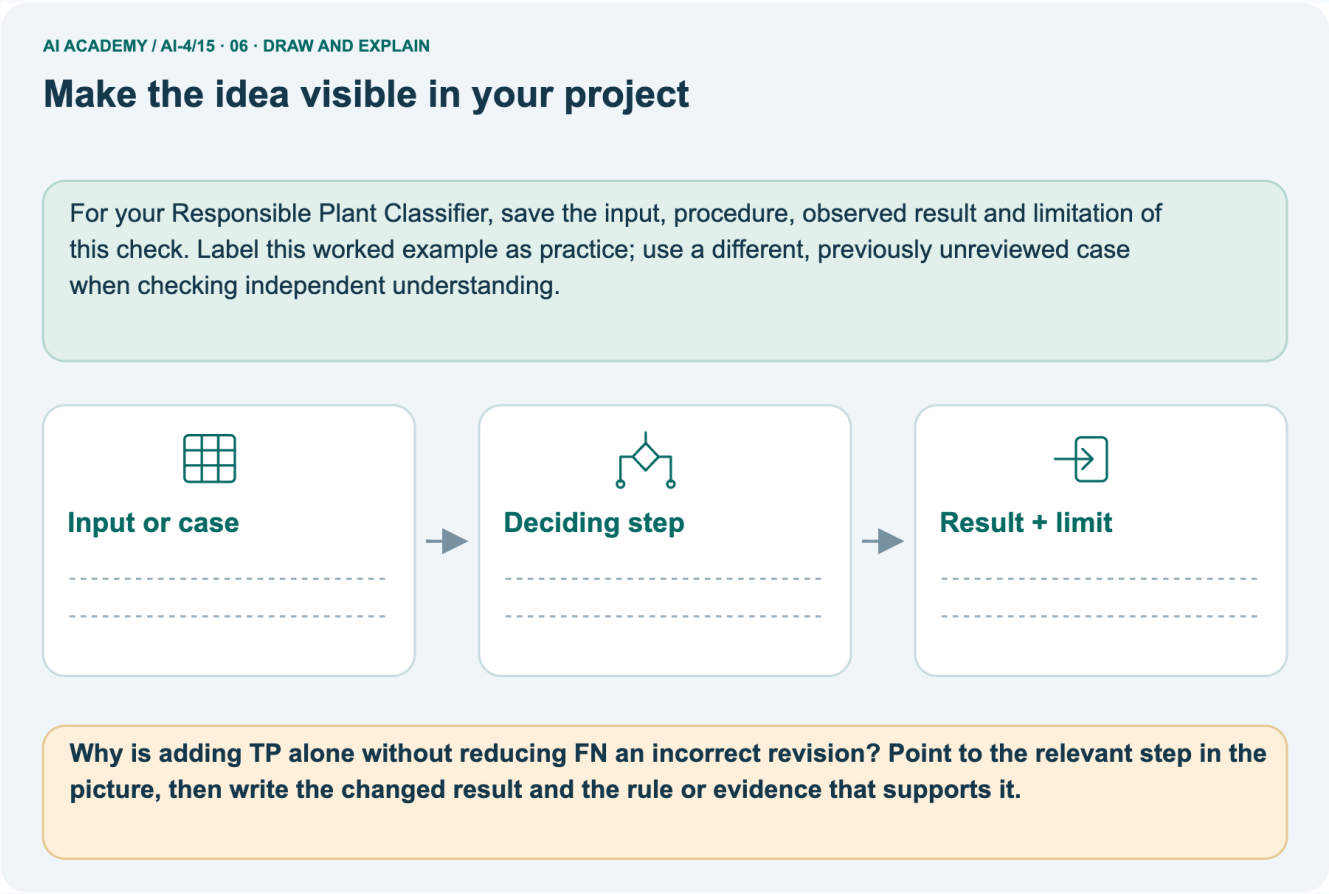
Read the picture in order. Correct=4+12=16 of total 20, giving**80%**. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: The matrix depends on positive-class definition and reference labels.

Accuracy alone hides the difference between misses and false alarms.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is adding TP alone without reducing FN an incorrect revision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.